

FRED ZHENGYU FU

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EDUCATION

University of Florida

Gainesville, United States

- **M.S. in Forest Resources and Conservation (thesis)**

May 2027 (expected)

GPA: 4.0/4.0

Advisor: Dr. Jiangxiao Qiu

Zhejiang University

Hangzhou, China

- **B.S. in Ecology**

Jun 2025

GPA: 3.3/4.0 | **Major GPA: 3.83/4.0**

Advisor: Dr. Shuang Liang and Dr. Weile Chen

Core Courses

University of Florida

- Analysis of Forest Ecosystem (100/100)
- Image Processing of Remote Sensing (95/100)
- Natural Resources Climate Change (94/100)
- Forest Ecosystem Health (97/100)
- Geospatial Application of UAS (94/100)
- Geospatial AI for Sustainable Science (92/100)

Zhejiang University

- Conservation Biology (94/100)
- Environmental Biology I (93/100)
- Field Ecology (91/100)
- Community and Ecosystem Ecology (91/100)
- Restoration Ecology (92/100)
- Environmental Biology II (92/100)
- Global Change Ecology (91/100)
- Population Ecology (90/100)

RESEARCH EXPERIENCE

University of Florida

A Knowledge-Guided Machine Learning Framework for Modeling Productivity and Carbon Dynamics in Florida Subtropical Pastures

Graduate Research Leader

Sep. 2024 – Present

- Develop a Knowledge-Guided Machine Learning (KGML) framework that couples a process-based model, deep learning, and Earth observations (remote sensing, eddy covariance, and field measurements) to predict pasture productivity and carbon dynamics.
- Built a grazing module in APSIM to more realistically represent grazing effects on pasture growth and carbon allocation.
- Collected and integrated soil, weather, and management data from multiple sources into model-ready inputs for site initialization and parameterization.
- Processed field measurements, remote sensing products, and flux tower observations for model calibration and cross-site validation.
- Training neural networks on synthetic data generated by the calibrated process-based model, with physical constraints embedded in the learning process to improve accuracy and transferability.

Responses of greenhouse gas fluxes to altered precipitation, land management, and cattle grazing

Research Assistant

Sep 2024 – Present

- Conduct fieldwork in a Florida Long-Term Agroecosystem Research (LTAR) station, focusing on soil gas flux measurements and soil sampling.
- Perform laboratory analyses of soil and gas samples.
- Support the coordination of routine field campaigns, including logistics, sample handling, and communication among research team members.

Agricultural land-use intensification effects on soil carbon, nitrogen, and nutrient stoichiometry across a subtropical rangeland

Research Assistant

Feb 2025 – Present

- Conduct root sampling under different experimental treatments in LTAR station.
- Wash, sort, and analyze root samples to quantify root morphology and trait responses.

Zhejiang University

Modeling the Relationship between Tree Species Diversity in Forest Plantations and Carbon Sequestration Capacity under Projected Climate Change Scenarios

Project Leader, Advised by Dr. Shuang Liang and Dr. Weile Chen

May 2023 – May 2024

- Synthesized species-specific ecological and life-history traits from published literature and public databases to support the parameterization of multiple tree species in LANDIS-II.
- Analyzed LANDIS-II simulations to compare forest productivity between monoculture and mixed-species plantations across different stand ages.
- Evaluated how stand age and species composition influenced the productivity differences between monoculture and mixed-species plantations and developed corresponding forest-management recommendations.

Integrated Field Course of Biology

Group Leader, supervised by Dr. Jun Chen and Dr. Yunpeng Zhao

Jun 2022 – Jul 2022

- Conducted ecological field investigations including vegetation surveys and environmental measurements.
- Integrated field data for ecological analysis and interpretation.

CONFERENCES & PUBLICATIONS

- **Fu, F.**, Wu, R., Cha, J.-Y., Gedert, J., Boughton, E., Martens-Habbena, W., & Qiu, J. A Knowledge-Guided Machine Learning Framework for Modeling Productivity and Carbon Dynamics in Florida Subtropical Pastures. AGU Fall 2026. (*Abstract submitted; Manuscript in preparation*)
- Wu, R., Boughton, E. H., **Fu, F.**, Martens-Habbena, W., Sonnier, G., & Qiu, J. Agricultural land-use intensification effects on soil carbon, nitrogen, and nutrient stoichiometry across a subtropical rangeland. *CANVAS Fall Meeting 2026*. (*Abstract submitted*)
- Cha, J.-Y., Diaz, A., Gedert, J., Brown, S., Gaudreault, B., Wu, R., **Fu, F.**, Siclare, W., Boughton, E. H., Qiu, J., & Martens-Habbena, W. Do Soil Microbial Processes and Environmental Conditions Interact to Shape Greenhouse Gas Fluxes in Subtropical Grasslands? AGU Fall 2026. (*Abstract submitted*)
- Cai, L., Cui, W., **Fu, F.**, Chen, W., Liang, S. Mixed-species plantations sustain long-term, climate-resilient carbon benefits over monocultures in subtropical China. (*Manuscript submitted*)

PROFESSIONAL DEVELOPMENT & LEADERSHIP

Selected Participant, International Summer school *Aug 2026, Beijing Normal University*
Frontiers in Telecoupling and Metacoupling Research for Addressing Global Sustainability Challenges

- Selected to participate in a competitive international summer school led by Prof. Jianguo (Jack) Liu, receiving intensive training in telecoupling and metacoupling frameworks for analyzing coupled human and natural systems across distances.

Co-developed a group research proposal applying the metacoupling framework to assess the effects of the EU Deforestation Regulation (EUDR) on global agricultural trade and land-use change.

Lab manager in Jiangxiao Qiu's Lab *Feb 2025 – Apr 2025, University of Florida*

- Coordinated monthly field campaigns with LTAR collaborators, including site confirmation, scheduling, logistics, and team assignments.

Managed field equipment and sample inventories to maintain sample traceability and support data quality.

Captain, Department Badminton Team *May 2022 – Jun 2024, Zhejiang University*

- Fostered an inclusive and motivating team environment by organizing training, supporting teammates of different skill levels, and encouraging broad participation in competition.
- leading the team to record-breaking results for two consecutive years and a fourth-place finish in the university-wide competition.

SKILLS & INTERESTS

Skills:

- Programming & Analysis: R, SAS, Python
- Process-based Model: LANDIS-II, APSIM
- GIS & Remote Sensing: ArcGIS, Google Earth Engine (GEE), ENVI 6.0
- Field & Lab Methods

Interests:

- Ecosystem / Agrosystem Modeling
- Artificial Intelligence
- Remote Sensing
- Global Change Ecology
- Landscape Ecology